

Algebra 1 — Homework 4

Source: Tutorial sheet 4 (Tutorials/Tutorial 4.pdf) Opened 3 Sep 2026 Due Mon 14 Sep 2026, 11:59 PM (eve of Test 1)

Submit: 2 (one of the four proofs, in your own words), 5 (computation only), 6(c), 7(a), 11(a) and (c). Tutorial 5 repeats: “finish HW 4 computations, especially 11(a),(c)”.

Justify all computations. A computer may be used to verify row reduction; setup and interpretation must be your own.

2. If S is a finite spanning set of a vector space V and I is linearly independent, then $|I| \leq |S|$. Give *one* of the four proofs from the sheet: (a) contradiction via a wide matrix acting on a row of abstract vectors; (b) contradiction by row-reducing the tall system; (c) reduction to $\mathbb{R}^n$ via an isomorphism; (d) Steinitz exchange lemma.

5 (computation). Let $A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{pmatrix}$. Show A is invertible, write A as a product of elementary matrices, and find A^{-1} .

6(c). Let $A = \begin{pmatrix} 2 & 3 & 1 \\ 1 & 2 & 3 \end{pmatrix}$. Show A has infinitely many right inverses and find them all. Then give a matrix B with infinitely many right inverses and find them all.

7(a). Let $v_1 = (1, 2, 1)^T$, $v_2 = (0, 1, 2)^T$, $v_3 = (2, 0, 1)^T$. Show $B = \{v_1, v_2, v_3\}$ is a basis of $\mathbb{R}^3$. Find the change of basis matrix P from the standard basis to B . Find w with $[w]_B = (4, 5, 5)^T$, and find $[v]_B$ for $v = (4, 5, 5)^T$.

11. Write formulas, with justification, for these maps $\mathbb{R}^2 \rightarrow \mathbb{R}^2$:

- (a) the linear map sending $(1, 2)^T \mapsto (3, 4)^T$ and $(2, 5)^T \mapsto (10, 13)^T$;
- (c) reflection in the line $y = kx$, done two ways: directly in the standard basis, and via a more convenient basis followed by change of basis. (Maybe name an angle.)